

VOID-NAV

An infrastructure-independent SOS network: public SOS nodes and a controlled LoRa relay mesh that carry prioritised, acknowledged emergency messages when towers and internet are down.

PROBLEM STATEMENT

In a major disaster, cellular towers, internet and power fail together. Survivors cannot ask for help, share their location, or reach family, even when responders are only a few kilometres away.

A replacement network must work with **no cellular or internet dependency**, and it must **not collapse under its own traffic**: as more survivors transmit, retransmissions, duplicates and simultaneous packets can saturate a low-bandwidth radio channel.

We need controlled forwarding, priority for urgent cases, delivery acknowledgements and location reporting, within LoRa's airtime limits.

Team: Team VOID Team ID: [to be filled] Category: Next-Gen Communication

Members: [Name 1] · [Name 2] · [Name 3] · [Name 4]

Towers down

no cellular, no internet

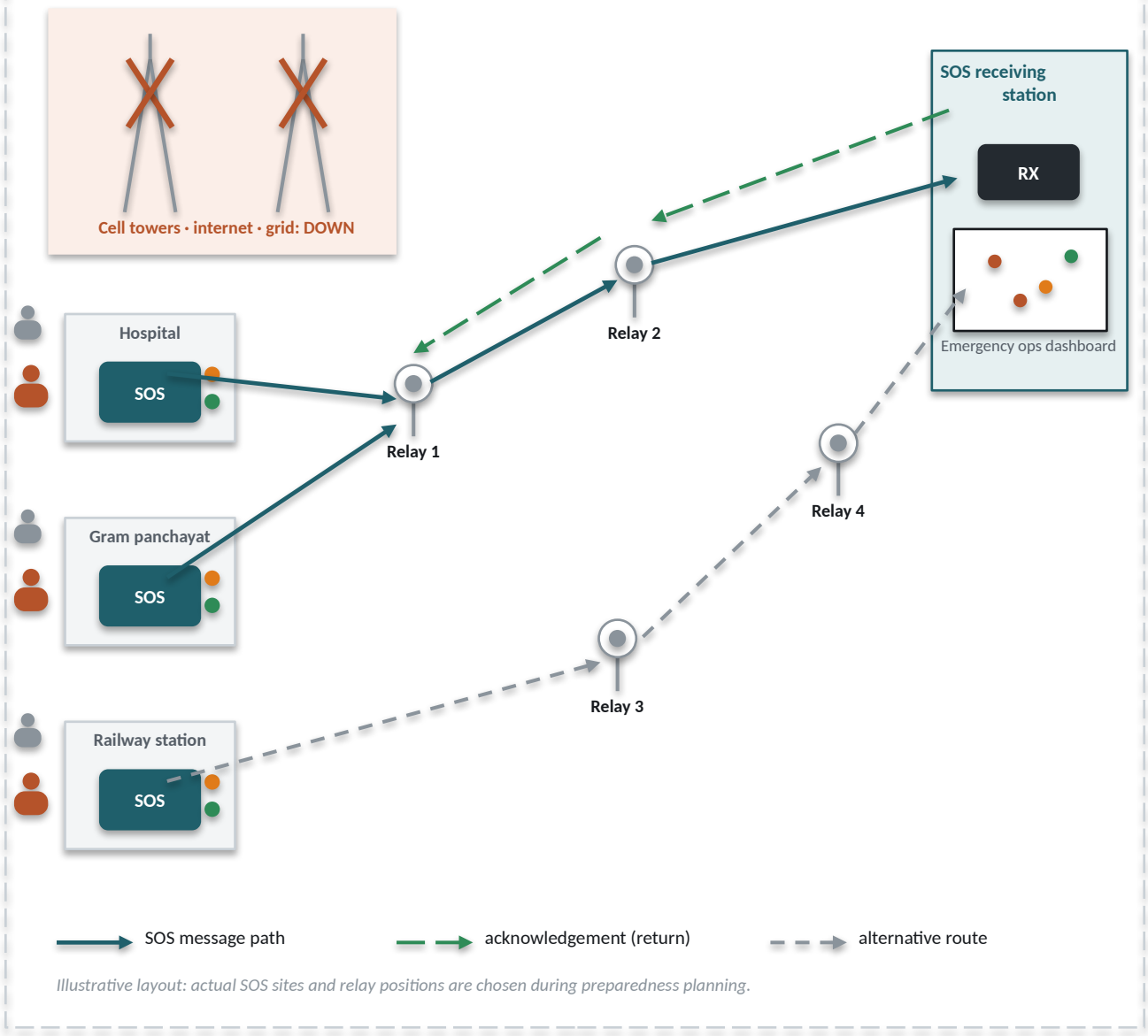
Shared channel

limited LoRa airtime

Must confirm

sent ≠ delivered

DISASTER ZONE → PUBLIC SOS NODES → RELAYS → RESPONSE



Proposed Solution

Public SOS points + structured requests + controlled LoRa relaying + end-to-end acknowledgements

PROBLEM AT HAND

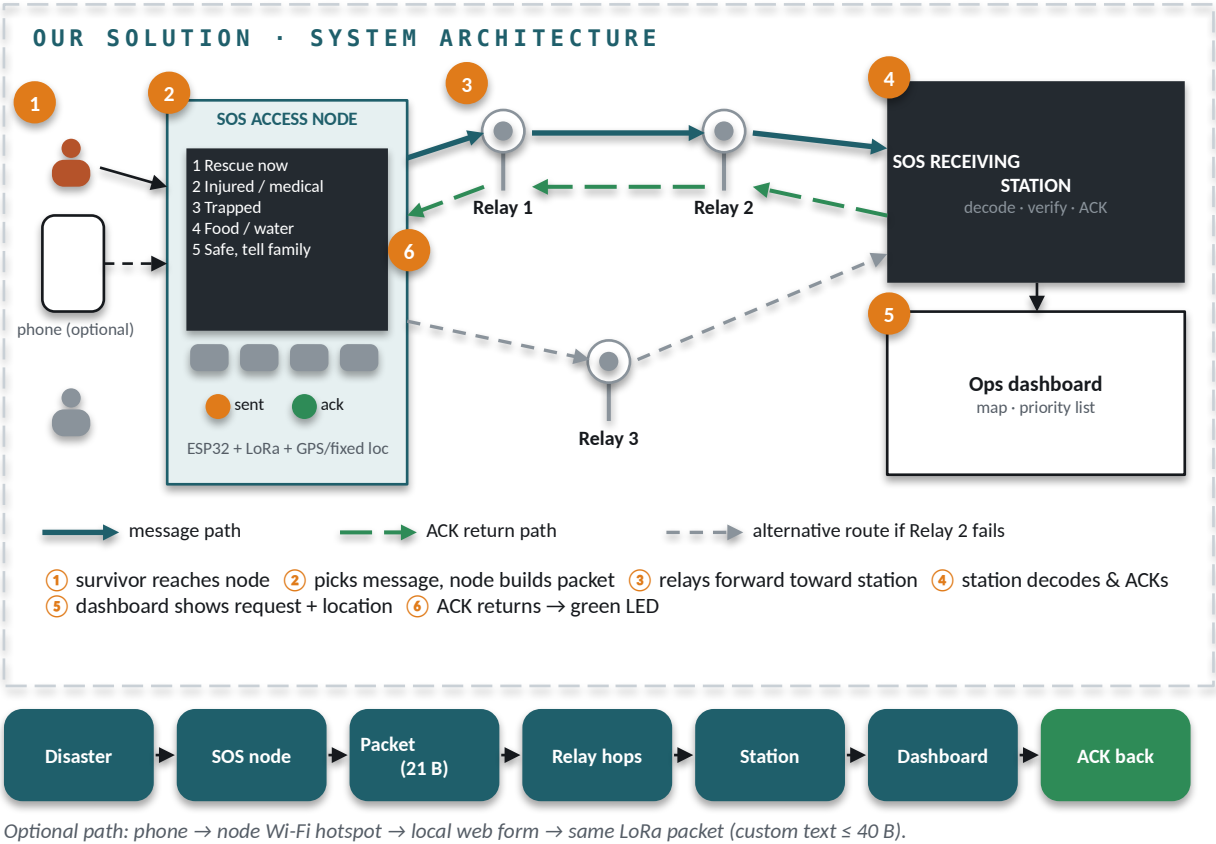
- 1

Communications fail
Towers, internet and power go down together; phones lose service.
- 2

No way to ask for help
Survivors can't reliably request rescue, medical aid or supplies.
- 3

Location is unknown
Families and responders can't tell where a request came from.
- 4

Forwarding congests
Unchecked rebroadcasts, duplicates and collisions waste scarce LoRa airtime.



KEY FEATURES

- Infrastructure-independent

local radio only; no cellular, internet or cloud
- Structured SOS menus

pre-loaded categories, yes/no and counts
- Multi-hop forwarding

relays carry packets beyond one radio link
- Phone-to-node Wi-Fi (opt.)

local web form for short custom text
- Location-linked reports

node GPS or surveyed fixed-node location, labelled
- Two-way acknowledgement

orange = sent · green = delivered
- Priority, congestion-aware

P0 rescue first; dedup, hop limit, backoff
- Store-and-forward

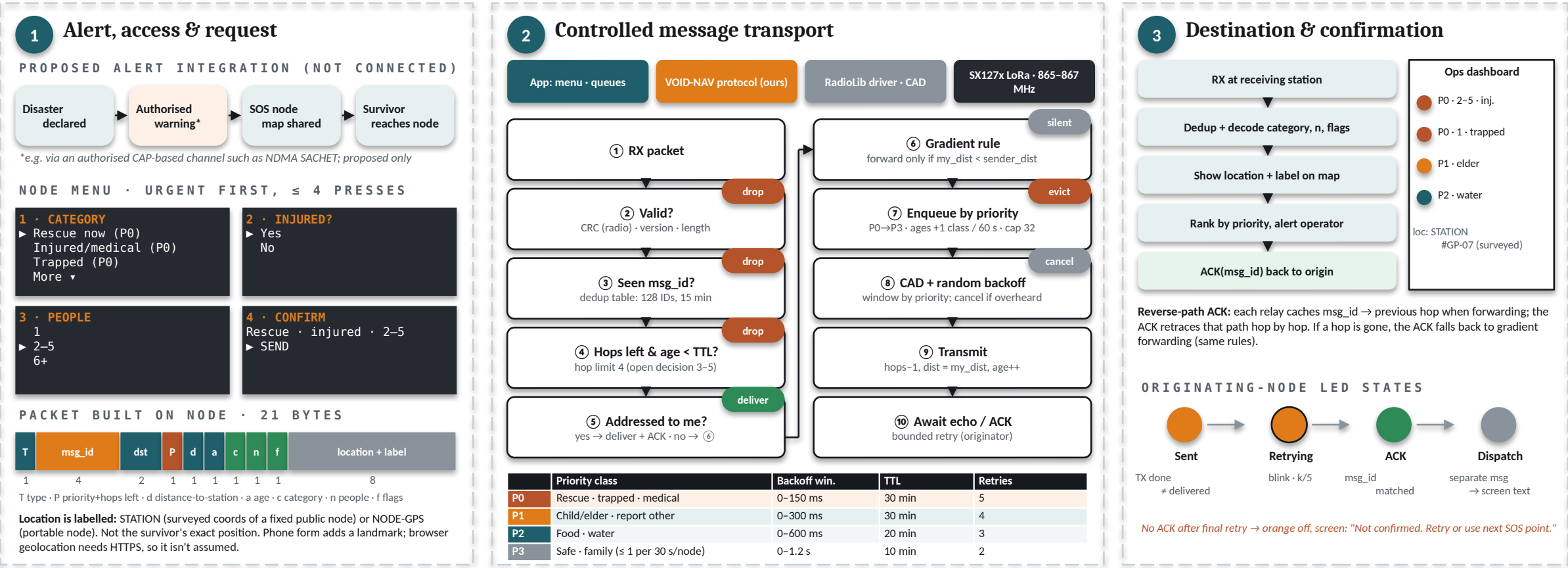
buffer with expiry during link outages

WHY WE STAND OUT

LoRa itself isn't the novelty. The value is the **integration**: accessible public SOS points + one-touch structured requests that fit in ~21 bytes + a forwarding protocol we implement ourselves (**distance-gradient forwarding, duplicate suppression, hop limits, priority queues, bounded retries**) + **end-to-end acknowledgements** that tell a survivor their message actually arrived, with explicit, honest location labels.

Technical Approach

How a request is created, carried across a congested shared channel, and confirmed back to the survivor



Feasibility, Viability & Risk Mitigation

Built incrementally on commodity ESP32 + LoRa hardware, with every risky assumption named and testable

A Why the system is feasible

- ✓

ESP32 controller
dual-core MCU with built-in Wi-Fi; mature Arduino/ESP-IDF toolchains
- ✓

LoRa modules available
SX127x-family modules with 865–867 MHz variants are sold in India
- ✓

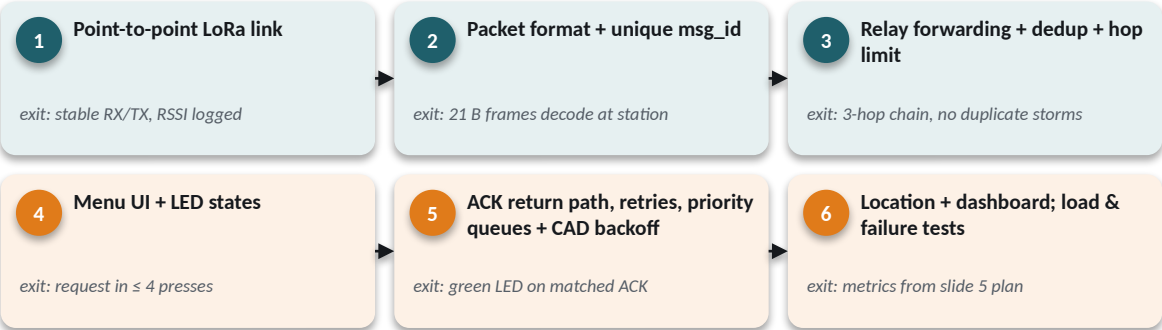
Local Wi-Fi interface
ESP32 SoftAP + small web server serves a form with no internet
- ✓

Small structured payloads
21 B request ≈ 185 ms airtime at SF9/125 kHz (formula-based)
- ✓

Incremental build
each layer (link → relay → ACK → priority) testable on its own
- ✓

No cloud, no cellular
the LoRa path is fully local; the dashboard runs on a laptop

B How we build it



C Risks & mitigation

Risk	Effect	Mitigation
Simultaneous TX / collisions	lost packets; retries add load	CAD before TX, priority-scaled random backoff, cancel if overheard; collisions reduced, never eliminated
Relay congestion	queues fill, latency grows	gradient forwarding (not blind flooding), P3 rate limit, queue cap 32 with lowest-priority eviction
Lost ACKs → repeats	duplicate traffic	retries reuse msg_id → relays drop dups; station re-ACKs without re-alerting; capped retries
Relay or station offline	route breaks	another lower-distance relay forwards; store-and-forward until TTL; distance beacons refresh
Wrong / missing location	help sent to wrong place	explicit label: STATION / NODE-GPS / NONE; surveyed coords at install; GPS fix age
Limited bandwidth & payload	text crowds out urgent traffic	21 B structured frames; custom text ≤ 40 B, single frame, no fragmentation in MVP
Forged / spam messages	false alarms, wasted airtime	node-ID allow-list + rate limits (MVP); message authentication with a network key (proposed)
Power failure at SOS site	node offline	battery + solar enclosure (proposed); relay sleep; battery level in beacons
Radio regulation	non-compliance	stay in 865–867 MHz exempt band, 125 kHz BW (< 200 kHz); verify power limits in GSR 853(E)

D Prototype vs deployment

Capability	Hackathon PoC	Deployment
LoRa P2P + 3-hop relay	planned demo	field test
Dedup, hop limit, gradient rule	planned demo	tune at scale
Priority queues + CAD backoff	planned demo	load test
End-to-end ACK + LEDs	planned demo	field test
Phone Wi-Fi form	optional	optional
Fixed / GPS location label	planned demo	survey all sites
Ops dashboard (offline)	planned demo	multi-station
Solar enclosure, auth key	—	proposed
Public-warning integration	—	proposed, needs partner

Status: "planned demo" = target for our hackathon build, [confirm before submission]. Nothing above is claimed as already tested.

BOM template: ESP32 · LoRa module + antenna · OLED + 4 buttons · 2 LEDs · GPS (portable) · battery/solar · enclosure → unit costs [fill from supplier quotes].

Impact & Benefits

Keeping a usable emergency pathway open when conventional infrastructure is gone

**Survivors without signal**
can request help from a nearby public SOS point

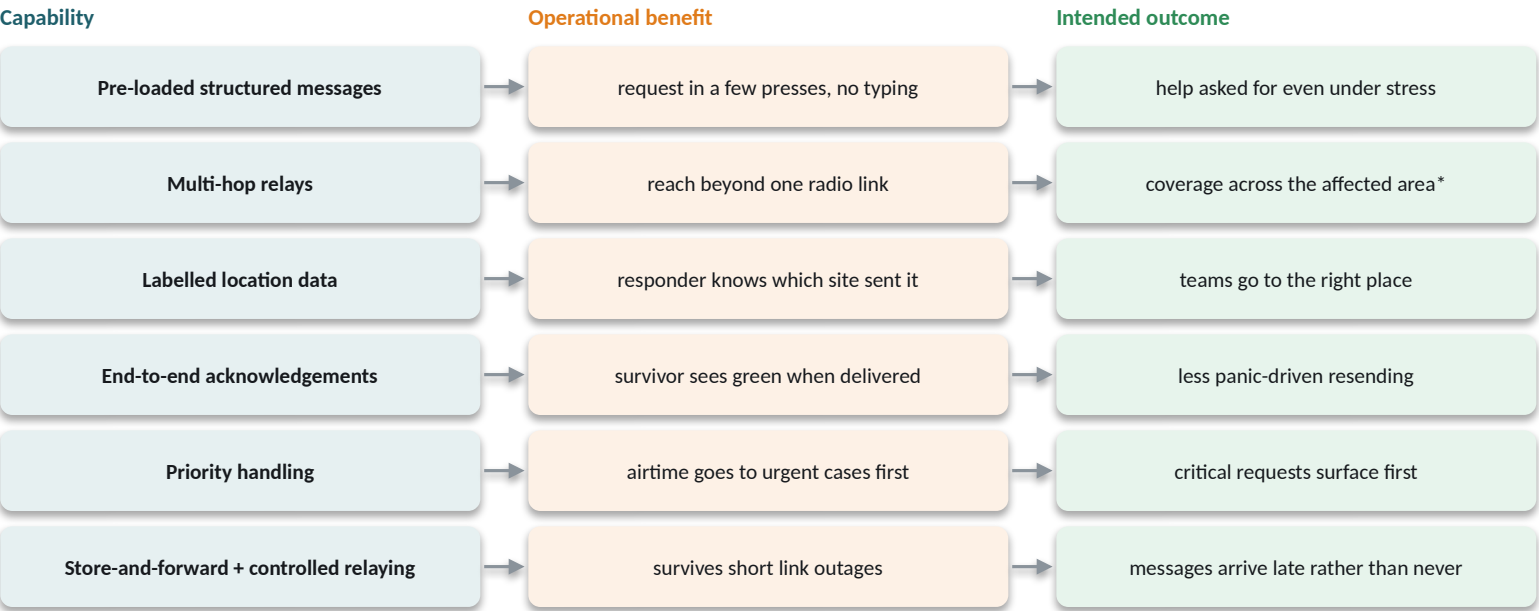
**Urgent medical / rescue**
P0 requests get first use of scarce airtime

**Families**
"I am safe" messages travel when calls cannot

**Authorities**
one prioritised, location-labelled list of requests

**Public facilities**
hospitals, panchayats, stations become comms points

CAPABILITY → OPERATIONAL BENEFIT → INTENDED OUTCOME



*subject to radio range, terrain, relay placement and channel capacity, which must be measured on site.

CAPACITY & LATENCY (ESTIMATES)

More relays ≠ more capacity. Nodes in radio range share one channel. One P0 request over 3 hops plus its ACK uses ≈ 3×185 + 3×144 ≈ 1.0 s of airtime, so at a cautious 20–30 % channel load one shared channel carries roughly 12–18 such requests per minute. Scaling means more receiving stations and separate areas, not just more relays.

Per hop, P0 (SF9, 21 B)	ms
CAD check (~2 symbols)	≈ 8
Random backoff (avg of 0–150)	≈ 75
Time on air	≈ 185
Processing / queue	≈ 10
3 hops forward	≈ 0.85 s
ACK back, 3 hops (10 B)	≈ 0.7 s
Round trip, idle channel	≈ 1.6 s

Proposed targets (to measure): P0 ACK ≤ 5 s at 3 hops · delivery ratio ≥ 95 % at test load · transmissions per delivered request ≤ hops + 2 · reported at 1, 5, 20 requests/min and with one relay removed.

Intended outcome: when towers and internet are gone, survivors still have a nearby place to ask for help, and responders still receive a prioritised, location-labelled list of who needs it.

Research & References

What published work establishes, how it shapes VOID-NAV, and what we still have to prove

Area	Established technique / finding (click to open)	How it informs our design	Status
LoRa airtime	Semtech AN1200.13, LoRa Modem Designer's Guide: time-on-air formula; airtime rises with SF and payload ↗ link	21 B structured frame; SF9 baseline ≈185 ms; custom text capped at 40 B	Design basis
LoRa scalability	Bor, Roedig, Voigt, Alonso. "Do LoRa Low-Power Wide-Area Networks Scale?" ACM MSWiM 2016: collisions limit density; parameter choice and multiple sinks help ↗ link	short frames, priority backoff, plan for several receiving stations	Needs evaluation
Managed flooding	Meshtastic, Mesh Broadcast Algorithm: hop-limited flooding, listen-before-rebroadcast, SNR-based contention window ↗ link	dedup, hop limit, cancel-if-overheard; we add a distance-gradient rule and priority classes	MVP core
Why flood routing	Meshtastic blog, "Why Meshtastic uses managed flood routing": trade-offs of flooding vs routed meshes ↗ link	keep routing simple; limit flood scope with gradient + hop limit	MVP core
Channel activity detection	Semtech, "Channel Activity Detection: ensuring your LoRa packets are sent": CAD detects LoRa preambles/chirps, not full carrier sense ↗ link	CAD + random backoff before TX; collisions still possible	MVP (to test)
SX1276 radio	Semtech SX1276/77/78/79 datasheet: LoRa modem, CAD mode, programmable SF/BW/CR ↗ link	radio settings and CAD use match the actual chip	Design basis
ESP32 SoftAP	Espressif ESP-IDF Wi-Fi API (AP/STA modes); SoftAP example: max_connection default 4 ↗ link	optional phone form; few phones per node at a time	Optional
Browser location	MDN Geolocation API & Chrome: geolocation only in secure contexts (HTTPS) ↗ link	don't assume phone GPS on a local HTTP portal; use labelled node location	Design constraint
Indian band rules	DoT clarification on 865–867 MHz (1 W TX, 4 W ERP, 200 kHz carrier); GSR 853(E) 2021, 865–868 MHz SRD exemption rules ↗ link	125 kHz BW inside 200 kHz; verify current limits in the 2021 gazette text	Must verify
Regional plan	LoRa Alliance RP002-1.0.3 Regional Parameters, IN865-867 section ↗ link	channel-plan reference (we do not run LoRaWAN)	Reference
Public warning	NDMA CAP-based Integrated Alert System (SACHET), built with C-DOT ↗ link	proposed route to announce SOS node locations; no integration exists	Proposed

Read this right: published work supports individual techniques (airtime, CAD, managed flooding, band rules). It does not show that VOID-NAV as a whole works. Our own contribution is the integration and protocol design, and every performance figure in this deck is an estimate or target until measured on our prototype.